

Accessing Leiden via bus



Jeroen Ockers

Universiteit Leiden
May 28th, 2025



**Universiteit
Leiden**
The Netherlands

Table of Contents

Introduction

Applying the algorithm

Next steps

Make Leiden more accesible

- ▶ Less mobile people
- ▶ Encourage bus use

- ▶ Origin & Destination points
- ▶ Small increase in cost

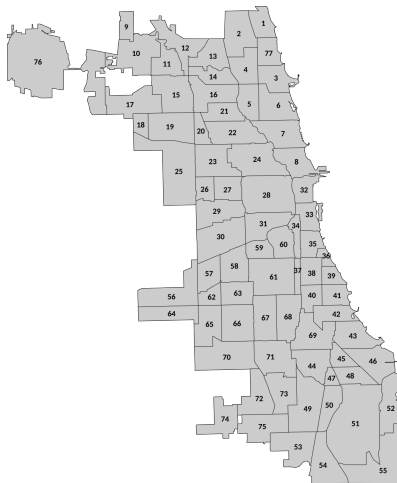


Figure: Chicago community areas

Applying the algorithm

Data required:

- ▶ Destinations

Applying the algorithm

Data required:

- ▶ Destinations
- ▶ Origins

Applying the algorithm

Data required:

- ▶ Destinations
- ▶ Origins
- ▶ Busstop locations

Applying the algorithm

Data required:

- ▶ Destinations
- ▶ Origins
- ▶ Busstop locations
- ▶ Current bus schedule

Applying the algorithm

Data required:

- ▶ Destinations
- ▶ Origins
- ▶ Busstop locations
- ▶ Current bus schedule
- ▶ Current travel data

Destinations

► GP locations

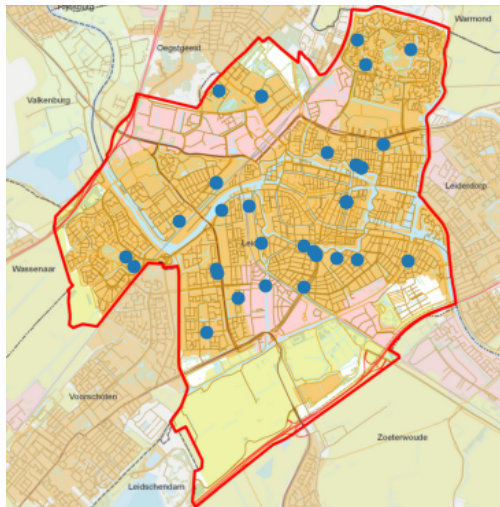


Figure: GP locations in Leiden

Destinations

- ▶ GP locations
- ▶ Quality

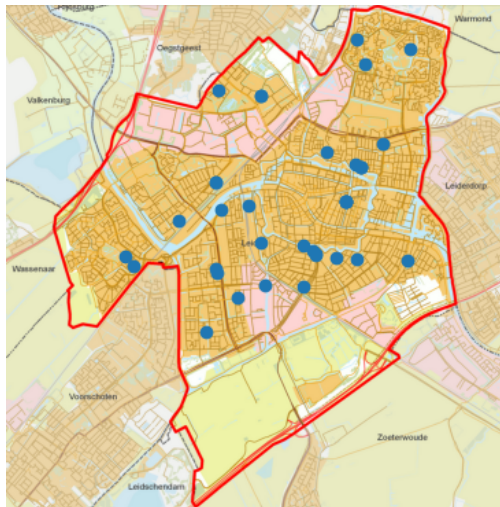


Figure: GP locations in Leiden

Destinations

- ▶ GP locations
- ▶ Quality
- ▶ No hospitals

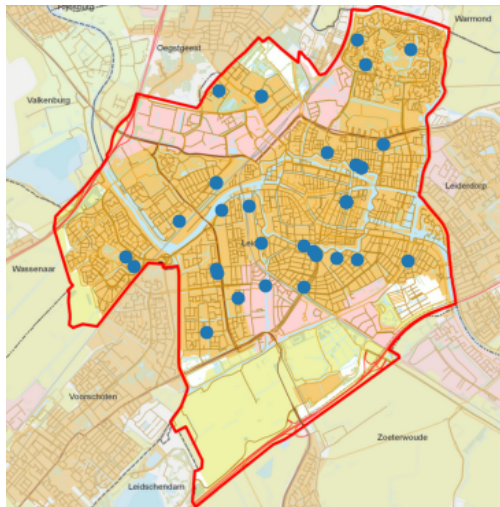
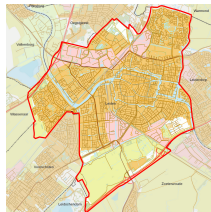
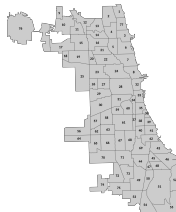


Figure: GP locations in Leiden

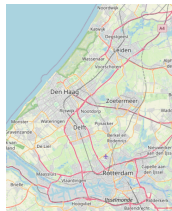
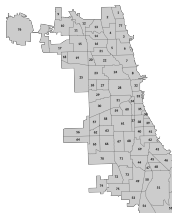
Origins

- Chicago regions too large



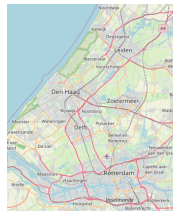
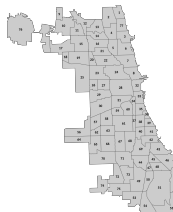
Origins

- Chicago regions too large



Origins

- ▶ Chicago regions too large
- ▶ Two methods



► Postcode

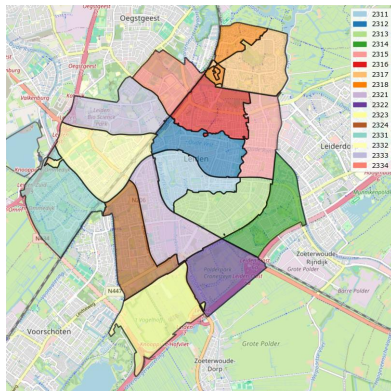


Figure: The different postcode regions of Leiden

- Postcode
- Origin points

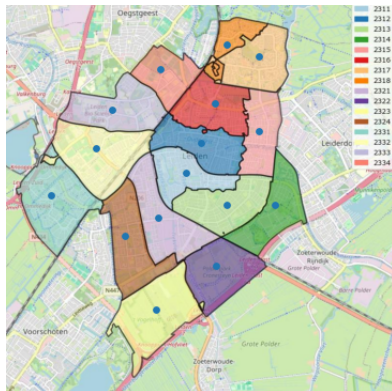


Figure: The different postcode regions of Leiden

- ▶ Postcode
- ▶ Origin points
- ▶ Population known

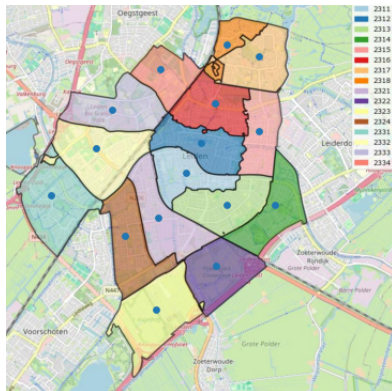


Figure: The different postcode regions of Leiden

- ▶ Postcode
- ▶ Origin points
- ▶ Population known
- ▶ Still quite large

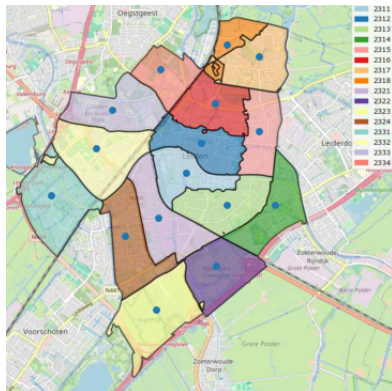
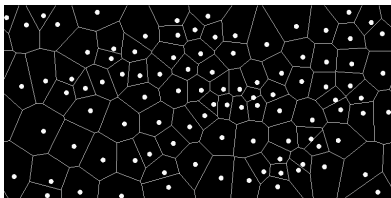


Figure: The different postcode regions of Leiden

Voronoi

- ▶ Weight points
- ▶ Shortest distance



► Busstops

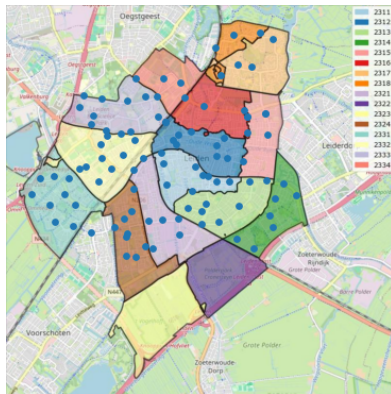


Figure: Busstops in Leiden

Voronoi

- ▶ Busstops
- ▶ Approximate populations
- ▶ Trips start at busstop

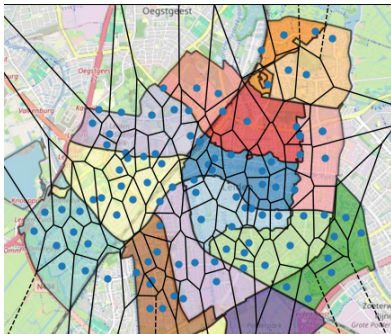


Figure: Busstops in Leiden

Next steps

- ▶ Travel data

Next steps

- ▶ Travel data
 - ▶ Acquire

Next steps

- ▶ Travel data
 - ▶ Acquire
 - ▶ Approximate

Next steps

- ▶ Travel data
 - ▶ Acquire
 - ▶ Approximate
- ▶ Run on PC4 regions

Next steps

- ▶ Travel data
 - ▶ Acquire
 - ▶ Approximate
- ▶ Run on PC4 regions
- ▶ Estimate voronoi populations

Next steps

- ▶ Travel data
 - ▶ Acquire
 - ▶ Approximate
- ▶ Run on PC4 regions
- ▶ Estimate voronoi populations
- ▶ Run on voronoi regions